Section A

Cluster 1: Development, Economy and Environment

1	Resource 1 shows the distribution of states in India with solar power projects in 2017. Resource 2 is an article on use of solar energy in India. Resource 3 shows China's plans for a 'super dam' with existing dams on the rivers. Resource 4 shows Sustainable Development Goal (SDG) indicator 7.2.1 on share of final energy use that comes from renewable sources in 2020.		
	(a)	Explain how provisioning services can be classified based on their renewability.	**[4]**
		Award 1 mark for each explanation of provisioning services classified based on their renewability. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • Non-renewable resources, also known as stock resources, are resources that have taken millions of years to form and so their availability is finite. o There is no possibility of them being replenished on a timescale of relevance to human society. • Renewable resources are resources that are naturally renewed within a sufficiently short timespan to be of use to human society. • Renewable resources can also be classified into 2 categories, critical and non-critical zones. o Resources in non-critical zones have continued availability that is independent of actions of society while resources in critical zones have continued availability is dependent upon management by society • It is possible for renewable resources to become non-renewable if they are mismanaged and their regenerative capacity is exceeded. • Resources can also be classified along a continuum where all resources are renewable on some time scale o what matters for the sustainability of future supplies is the relative rates of replenishment and use. **AO1**	
	(b)	With reference to Resource 1, describe the distribution of solar power projects in India.	**[4]**
		Award 1 mark for each description of the distribution of solar power projects in India. Award a maximum of 1 additional mark for further development of each description, where applicable.	

		Possible answers include: • Solar power projects can be found in most states in India / 14 out of 18 states • However, some states (4) in the east and 1 state in the north and south of India do not have solar power projects. • The south has more solar power generated by projects in Tamil Nadu, Karnataka and Andhra Pradesh with projects 1.0GW and above. • Highest solar power generated in Telangana with more than 2.0GW generated. • There are more states with solar power projects installed that generates 1.0GW or less of solar energy • Among states with solar power generated, Delhi has the lowest solar energy generated with about 0.03 gigawatts. • Jharkhand as the anomaly being the only state in the east of India with solar power projects. **AO2**	
	(c)	With reference to Resource 2, explain the trade-offs associated with the use of solar power.	**[4]**
		Award 1 mark for each explanation of each benefit or limitation that lead to trade-offs. Award a maximum of 1 additional mark for each accompanying benefit/limitation that lead to trade-offs. Award a maximum of 2 marks for benefits or limitations given. Possible answers include: **Environmental trade-offs** • [Benefit] The use of solar power in India 'help to dampen the environmental and climate impacts as their economies expand' which help to reduce the nation's environmental footprint by replacing the use of fossil fuels with solar energy. ○ Reducing emissions reduces their negative impacts of climate change that the country might experience • [Limitation] Lead is released into the environment as 'recycling of lead-acid batteries in India is also not adequately regulated'. ○ When lead is released into the environment it can have negative impacts on wildlife and vegetation. **Social trade-offs** • [Benefit] The citizens in developing locales with unstable access to electricity experience socioeconomic benefits from solar power expansion, given that	

		'many villages get electricity for only two or five hours a day, and the power supply from the grid is erratic and unpredictable'. ○ Citizens might also get employment opportunities as a result of solar power expansion as access to electricity can support the expansion or provision of more business opportunities. • [Limitation] Prolonged exposure to lead can result in negative health impacts which include 'higher risks of cardiovascular disease and high blood pressure. The heavy metal can also cause damage to the kidneys, reproductive system, and the central nervous system'. **AO2**	
	(d)	With reference to Resource 3, explain how China's super dam in Tibet can cause conflict between China and India.	**[6]**
		Award 1 mark for each explanation point on how China's super dam can cause conflict between China and India. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: **Actions affecting Quantity of Water** • As the upper course of the Brahmaputra River is in China, filling up the reservoirs of their dams would result in India (located downstream) to have a lower water supply. ○ This would affect their access to water from the river and their food security would be affected as a result if they rely on growing crops to produce their food. • As the scale of the hydropower dam is described to be massive, when water is released, flooding might occur downstream of the Brahmaputra River. **Actions affecting quality of water** • Water quality is closely linked to quantity: decreasing water quantity concentrates pollution, while excessive water quantity, such as flooding, can lead to contamination from overflowing sewage. **Actions affecting timing of water flow** • China might release water from the Medog irregularly when there is demand for hydropower production.	

		○ These irregular releases destroy natural seasonal flow variations that trigger natural growth and reproduction cycles in many fish species, impacting food security of the areas located downstream. **Presence of the dam as a barrier** • Presence of a large dam can disrupt migration cycles of fish, affecting food security and livelihoods • Sediments can also be blocked by the dam, resulting in the areas downstream (including India) to deprived of nutrient-rich sediment for agriculture **AO2**	
	(e)	Explain the strategies that can be implemented to mitigate potential conflict over transboundary water resources as seen in Resource 3.	**[6]**
		Award 1 mark for each explanation of strategies that can be implemented to mitigate potential conflict on transboundary water bodies. Award a maximum of 1 additional mark for further development of each explanation, where applicable. **Possible answers include:** • [International Water Agreements] International water agreements are legal documents that can ensure the maintenance of quality, quantity and time of flow of water for all riparian states. ○ Specific terms such as workable monitoring provisions, enforcement mechanisms, and specific water allocation provisions that address variations in water flow and changing needs can be included to reduce chances of dispute and conflict. • [Cooperative water management mechanisms] can anticipate conflict and solve disputes, provided that all stakeholders are included in the decision-making process and given the means (information, trained staff, and financial support) to act as equal partners. ○ On the international level, river basin commissions with representatives from all riparian states have been successfully involved in joint riparian water resources management. ○ Effective transboundary water management starts at the national level, where coordination and cooperation between different ministries and water related institutions is needed, as are sufficient financing and political commitment	

		○ Public participation is fundamental to maximize agreement, enhance transparency and decision making, create ownership and facilitate the acceptance and enforcement of decisions and policies. • [Sustainable water management] To maintain or improve the quantity and quality of transboundary rivers, strategies implemented require the cooperation of riparian states as the water in these rivers are a shared resource ○ Payment for Ecosystem Services has the potential to be an internalize environmental costs, broaden sources of finance and create incentives for environmentally friendly investments and behaviour. **AO1**	
	(f)	With reference to Resource 4, suggest possible reasons for the global variation in the share of energy use that comes from renewable sources.	**[6]**
		Award 1 mark for each explanation for the share of energy use that comes from renewable sources. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Physical] Countries which have higher percentages of energy from renewable energy sources have access to large supply of renewable sources of energy ○ For example, access to large rivers which show potential in harnessing energy for hydropower such as River Nile and the Amazon / geothermal power in Iceland • [Physical] Most countries with more than 50% energy use from renewable sources are in the tropics. ○ The geographical location of the countries also exposes them to less seasonal variability in climate and more dependable solar radiation • [Physical] Countries in Africa might have higher percentages of final energy use from renewable sources due to them having large land areas for solar farms. • [Economic] Africa might have a substantially higher percentage of final energy from renewable sources as it receives investments from China for their development of solar energy. • [Economic] Some countries with higher percentages of renewable energy sources are CHLDs (in Europe), they have the financial ability to fund renewable energy infrastructure. • [Political] The governments of the African countries might have been using renewable energy sources as a priority to generate energy which led to more solar farms being developed and renewable energy sources bring a large part of their energy use.	

		• [Political] Countries like USA are CHLDs with low share of energy use from renewable sources as their governments focus on economic growth and allow them to rely on the use of fossil fuels. • [Political] Countries in Europe with higher share of renewable energy sources might focus on sustainable development and encourage/mandate the use of renewable energy sources as much as possible. **AO2**	

Cluster 2: Tropical Environments

2	Resource 5 shows the climograph of Pathein, a city located within the Irrawaddy Delta in Myanmar. Resource 6 shows the summer and winter monsoon circulations in humid tropical Asia. Resource 7 shows the forecast of Tropical Cyclone Mocha in May 2023 by the Myanmar Information Management Unit. Resource 8 shows land cover of the Irrawaddy Delta in 1974 and 2021.		
	(a)	Describe the climatic characteristics of Pathein, Myanmar as seen in Resource 5.	[4]
		Award 1 mark for each description of the climograph of Pathein, Myanmar. Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses: • Pathein has a high mean annual temperature of about 26.2°C • Its annual temperature range is low/small at about 5°C o Temperature is lowest at about 24°C at the end and start of the year, which gradually rises to its peak of 29°C in April before decreasing and staying constant from July to September. It then has a slight increase in temperature to September before decreasing to 24°C at the end of the year • It experiences a high total annual precipitation of about 2330mm • Pathein has a distinct wet and dry season. Its wet season is from May to October with at least 180mm of rainfall while its dry season is from November to April with at most 60mm of rainfall • Pathein has a tropical monsoon (Am) climate **AO2**	

	(b)	Using Resource 6, account for the climatic characteristics of Pathein, Myanmar as shown in Resource 5.	[6]
		Award 1 mark for each explanation of the climatic characteristics of Pathein. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses • In the middle of the year, the onset of moist southwest monsoon winds seen in R6 brings about high amount of average rainfall of about 18mm per day to the region where Pathein is. This explains the higher monthly rainfall in the middle of the year seen in R5. o Southeast monsoon winds coming from the southern hemisphere which deflect to the right in the northern hemisphere forming southwest monsoon pick up moisture from the Indian Ocean and Bay of Bengal to bring high amount of rainfall to western coast of Myanmar • However, at the end and start of the year when the northern hemisphere experiences winter, Pathein instead receives dry offshore northeast winds (as seen in R6), resulting in it to experience little to no rainfall as seen in R5. o The ITCZ shifts south as well, resulting in Pathein to experience the effect of the subsiding limb of the Hadley Cell • Yet, due to the arrival of moist monsoon winds that bring high average daily rainfall, it results in its TAP to be high o The arrival of the rising limb of the Hadley Cell in the period of Apr-Jun also exacerbate the amount of rainfall that Pathein receives • The relatively low latitude of Pathein (16°N) results in it to receive higher concentration of insolation and less of the insolation to be lost via scattering / reflection due to shorter atmospheric distance in which insolation has to travel through, resulting in it to have high MAT • Monthly temperature changes with the movement of the overhead sun relative to Pathein's latitude. As the position of the overhead sun shifts from the equator to the tropic of Cancer from March to June, Pathein that is located in the middle experiences its peak monthly temperature in April. o As the overhead sun shifts to the tropic of Capricorn by the end of the year, Pathein then experiences lower monthly temperature **AO2**	
	(c)	With reference to Resource 7, describe the development of Tropical Cyclone Mocha.	[4]
		Award 1 mark for each description of the development of Tropical Cyclone Mocha.	

		Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses • Cyclone Mocha was observed to be developing on 12 May 2023 and it was forecasted to last till 16 May 2023 • It started to form South of Bay of Bengal and made landfall in Myanmar o It travelled northwards and then in the northeast direction • It was first observed on 12 May 2023 and by 12.30pm, it has grown into a Cat-1 cyclone. As it continues to develop across the Bay of Bengal, it increased to a Cat-3 cyclone on 14 May 2023. It made landfall as a Cat-4 cyclone on 15 May 00:30h before travelling further north of Myanmar as it decreases its intensity to a tropical storm **AO2**	
	(d)	Explain atmospheric and surface conditions necessary for the development of tropical cyclones, such as the one seen in Resource 7.	[5]
		Award 1 mark for each explanation of the climatic characteristics of Pathein. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses: • A huge body of warm water (Sea temperatures of (usually) temperatures beyond 26°C and deeper than 60m is ideal) is required to ensure that the air parcels above the sea are highly humid, and the atmosphere unstable, creating an area of low pressure towards which winds will converge and initiating its formation o This will also ensure that there is sufficient energy for the storm to maintain its intensity (drawing from the latent heat that is emitted via condensation). o Resource 7 shows the cyclone initiating over the Bay of Bengal during May when the water is sufficiently heated by the overhead sun • Sufficient Coriolis force is needed to ensure that the surface winds that converge towards the area of low pressure (created by the rapidly rising warm and moist air) are deflected (beyond 5-10°N/S of the equator); this eventually leads to the spinning of the winds around the area of low pressure – known as the 'eye' – as they rise and condense (due to contact with heat energy from the warm sea surface temperatures).	

		○ This occurs due to the Earth's rotation upon its axis, with tropical cyclones spinning anti-clockwise in the Northern Hemisphere and Clockwise in the Southern Hemisphere ○ R7 shows the cyclone forming at a latitude beyond 5-10°N • Low (or absence of) vertical wind shear (little (or no) change in direction or velocity of wind across altitudes) allows for the upward spiralling winds to form thick vertical wall of clouds perpendicular to surface of the earth ○ High vertical wind shear will disrupt the flow of heat and moisture, inhibiting the development of the tropical cyclone. • Divergence of air aloft in the upper layers of the troposphere so that the rising air currents within the cyclone will be pumped out and a low pressure maintained at the centre. ○ This strong upper level divergence allows winds to evacuate the system faster than they are drawn in at the surface **AO1**	
	(e)	With reference to Resource 8, explain how ecosystem services provided by the Irrawaddy Delta could be affected by the changes in land cover.	[6]
		Award 1 mark for each explanation the effect on ecosystem services provided by deltas due to human activities. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses • Resource 8 shows the decrease in mangroves south of the delta. The loss of mangroves (vegetation associated with deltas) would reduce the amount of storm protection and increase coastal communities' vulnerability to natural hazards. ○ Mangroves help to anchor fine-grained coastal sediments and protect the coast from storms. They reduce the height and energy of wind and swell waves passing through them, reducing their ability to erode sediments and to cause damage to coastal areas • The loss of mangroves also results in the loss of natural resources, such as the provisioning services that mangrove provides, which include fisheries, forest resources and medicines ○ Mangroves provide a unique habitat for many species of plants and animals. The removal of mangroves can threaten biodiversity and have knock-on effects on fisheries	

		• The conversion of mangroves to dry crops land also reduces the amount of carbon sink, especially with the high rates of carbon sequestration that mangroves have • Urban and built-up areas are observed in the Irrawaddy Delta in 2021 when it was absent in 1974. The increase in people living in delta and built-up area can result in an increase in groundwater extraction that could cause salinisation issue, affecting water quality and agriculture o Excessive groundwater extraction result in less river discharge and affects geomorphic processes that develop deltas. As salts accumulate, farmers may turn to using saltwater irrigation which leads to poor plant health and a decline in agricultural productivity • It can be observed that large part of the Irrawaddy Delta has been converted from dry crops to irrigated crops. The use of water to irrigate crop can possibly affect river discharge and its associated geomorphic processes for the development of deltas. The lack of natural sedimentary processes can result in erosion of deltas that harms coastal communities' lives and livelihood o Many people depend on the delta for its nutrient rich sediments for agriculture. The erosion of delta can threaten livelihood of people and potentially displace people as well **AO2**	
	(f)	Explain how deltas are formed.	[5]
		Award 1 mark for each explanation of the formation of deltas. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses • [Lowered energy at the river mouth] As a river enters the sea at its mouth, river velocity is greatly reduced. Due to lowered river energy, fluvial deposition becomes dominant, and sediments fall to the river bottom o [Initiation of delta formation] The slowing velocity of the river and the build-up of sediment allows the river to break from its single channel as it nears its mouth, forming a deltaic lobe which includes its distributaries • [Materials sorting] In a deltaic lobe, heavier, coarser material settles first. Smaller, finer sediment is carried farther downstream. • [Geomorphic processes including the action of wave and tides] The formation of a delta depends on the interaction between geomorphic processes (which influence the river flow and sediment supply) and the distribution of the river sediment by waves and tidal currents	

		• [River-dominated] When deltas are situated in an environment with little tide and wave action, it is categorised as being river dominated as fluvial processes play the main role in their formation. It tends to be elongated in nature. • [Wave-dominated] Some deltas experience strong waves that strike the delta front stronger than the carrying power of the river. This results in the waves to carry away all the river sediment, resulting in such deltas to be generally smaller than other types ○ Wave-dominated delta tend to have few distributaries and seldom protrude into the sea. They shrink over time and eventually disappear • [Tide-dominated] On the other hand, tide-dominated deltas form when there exists large range between the high and low tide that result in strong tidal currents that overpowers the channel flow. ○ Tide-dominated deltas tend to have wide delta fronts. They tend to resemble estuaries that are elongated in the direction of tidal movement.	

Section B

Cluster 1: Development, Economy and Environment

3	'The world's progress towards sustainable development is alarmingly insufficient.' (United Nations, 2024) Evaluate the validity of the statement.	**[20]**
	Possible Approaches: Candidates could approach the question by making a judgement on the world's progress to sustainable development through a consideration of the social, economic and environmental dimensions. Candidates could analyse the progress and challenges made in the areas of sustainable use of resources, renewable energy, extractive industries and the global economy with the use of the Sustainable Development Goals (SDGs). Candidates could also approach the question by making a judgement on whether progress is insufficient through a consideration of two or more case studies. Candidates could analyse the effectiveness and challenges of using the targets and indicators to achieve the SDGs. *Levels marked using Generic Level Descriptors for 20m H2 essays* **AO3**	
4	'States have more influence on economic activities than labour.' Evaluate the validity of the statement above.	**[20]**
	Possible Approaches: Candidates could approach the question by making a judgement on whether states have more influence on economic activities through a consideration of the relative significance of states vis-à-vis labour. Candidates could analyse the influence of states on economic' operations through their role as regulators of economic activities and the degree to which states can exert an influence over the operations of firms such as TNCs. Candidates could also approach the question by making a judgement on whether states have a greater influence on economic activities than labour in some contexts than others through a consideration of two or more case studies. Candidates could analyse the contextual factors such as the nature of the economic activities and the motivation of states in trying to influence them, which may have contributed to the differences in the influence levels of states. *Levels marked using Generic Level Descriptors for 20m H2 essays* **AO3**	

Cluster 2: Tropical Environments

5	'The effects of fluvial floods vary.' Evaluate this statement.	**[20]**
	Possible Approaches: Candidates could approach this question by making a judgement on whether the effects of fluvial floods vary across places and different groups of people through a consideration of the factors that influence the extent of impact. Candidates could evaluate whether variations in the effects of fluvial floods exist due to physical (e.g. causes and characteristics of floods, geographical locations) and economic (e.g. development of countries, vulnerability, capacity to cope) factors that can influence the extent of effects of impacts. Candidates could analyse the relative influence of these factors in affecting the effects of fluvial floods, which include positive and negative socio-economic and environmental effects. Candidates could also approach the question by making a judgement on whether the effects of fluvial floods vary through a consideration of two or more case studies. Candidates could also consider the contextual factors such as the nature of the flood and vulnerability of the population which may have contributed to the variations in the extent of impact of floods or not. *Levels marked using General level Descriptors for H2 essays.* **AO3**	
6	Evaluate the factors influencing the formation of karst landforms in the humid tropics.	**[20]**
	Possible Approaches: Candidates could approach the question by making a judgement on whether the formation of karst landforms (cockpit karst, tower karst, caves) is influenced by various factors through a consideration of the relative roles of climate, vegetation, geology and human activities. Candidates could evaluate the roles of these factors by referring to contrasting locations, differing scale or different time scales. Candidates could analyse natural factors such as climate (temperature, precipitation, vegetation) in initiating the formation of karst landforms and geological characteristics that provide the necessary conditions for geomorphic processes in the formation and influencing the type of resultant landforms. The rise in human activities can modify these conditions but they play a secondary role in the formation of karst landforms. Candidates could also approach the question by making a judgment on whether some factors have a greater influence in some contexts than others through a consideration of	

	two or more case studies. Candidates could evaluate the relative significance of the different factors in each case study and analyse the interplay between the factors. *Levels marked using General level Descriptors for H2 essays.* **AO3**	